WORKED SOLUTION

STEP III 2002, Question 9

Mechanics • Inclined plane

STEP III 2002, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q9

The key idea

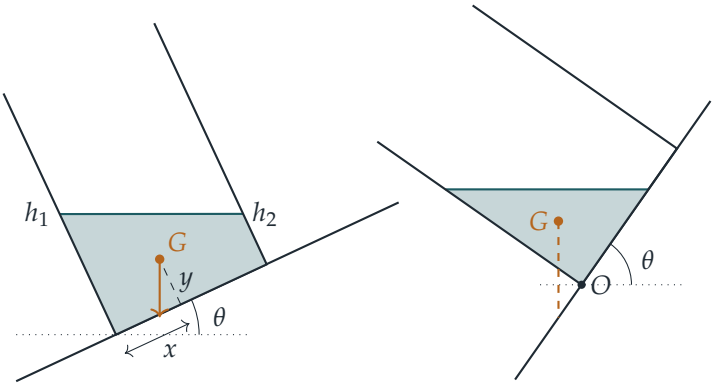
The free surface of the fluid is horizontal, so in the frame of the container (which is tilted at θ) it is a straight line sloping at angle θ to the base. A cross-section through the fluid is therefore a trapezium if the surface meets both side walls, and a right-angled triangle if it reaches the base before the upper wall. Finding the centre of mass is then a two-dimensional centroid calculation. For toppling, the container is light, so everything depends on where the vertical through the fluid's centre of mass meets the plane.

Solution

Setting up

We model the container as a light prism with walls of negligible thickness, the fluid as a uniform continuum at rest relative to the container, and the plane as rough enough to prevent slipping. Each vertical cross-section perpendicular to the horizontal base edges is the same, and the fluid is uniform, so the centre of mass of the fluid is the centroid of one cross-section. That cross-section has width a and area $ka^3/a = ka^2$.

Measure X along the base from the lower side wall (up the slope) and Y perpendicular to the base. The base rises at angle θ to the horizontal as X increases, so the horizontal surface falls at angle θ relative to the base: its height above the base drops by $\tan\theta$ for every unit of X. Write $d = a\tan\theta$ for the total drop across the width.



The left picture shows the case $k > \frac{1}{2}\tan\theta$ (the fluid touches both walls); the right shows $k < \frac{1}{2}\tan\theta$ with $\theta > 45°$, where the vertical through G misses the base.

The case $k > \frac{1}{2}\tan\theta$

Let the depths at the lower and upper walls be h_1 and h_2. Then $h_1 - h_2 = d$, and the trapezium has area $\frac{1}{2}a(h_1 + h_2) = ka^2$, so $h_1 + h_2 = 2ka$. Hence

$$h_2 = ka - \tfrac{1}{2}d = a(k - \tfrac{1}{2}\tan\theta),$$

which is positive exactly when $k > \frac{1}{2}\tan\theta$. So this is the case where the surface meets both walls.

Split the trapezium into a rectangle $a \times h_2$ and, on top of it, a right-angled triangle with legs a (along $Y = h_2$) and d (up the lower wall).

- Rectangle: area ah_2, centroid $\left(\frac{1}{2}a,\ \frac{1}{2}h_2\right)$.
- Triangle: vertices $(0, h_2)$, (a, h_2), $(0, h_2 + d)$, so area $\frac{1}{2}ad$ and centroid (the average of the vertices) $\left(\frac{1}{3}a,\ h_2 + \frac{1}{3}d\right)$.

Taking moments about the lower wall, with total area ka^2,

$$ka^2\, x = ah_2 \cdot \tfrac{1}{2}a + \tfrac{1}{2}ad \cdot \tfrac{1}{3}a \quad\Longrightarrow\quad kx = \tfrac{1}{2}h_2 + \tfrac{1}{6}d = \tfrac{1}{2}ka - \tfrac{1}{4}d + \tfrac{1}{6}d = \tfrac{1}{2}ka - \tfrac{1}{12}d.$$

Dividing by ka and using $d = a\tan\theta$,

$$\frac{x}{a} = \frac{1}{2} - \frac{\tan\theta}{12k}.$$

Taking moments about the base,

$$ka^2\, y = ah_2 \cdot \tfrac{1}{2}h_2 + \tfrac{1}{2}ad\left(h_2 + \tfrac{1}{3}d\right),$$
$$ka\, y = \tfrac{1}{2}h_2^2 + \tfrac{1}{2}dh_2 + \tfrac{1}{6}d^2 = \tfrac{1}{2}h_2(h_2 + d) + \tfrac{1}{6}d^2.$$

Now $h_2 = ka - \frac{1}{2}d$ and $h_2 + d = ka + \frac{1}{2}d$, so $h_2(h_2 + d) = k^2a^2 - \frac{1}{4}d^2$ and

$$ka\, y = \tfrac{1}{2}k^2a^2 - \tfrac{1}{8}d^2 + \tfrac{1}{6}d^2 = \tfrac{1}{2}k^2a^2 + \tfrac{1}{24}d^2.$$

Dividing by ka^2,

$$\frac{y}{a} = \frac{k}{2} + \frac{\tan^2\theta}{24k},$$

as required.

The case $k < \frac{1}{2}\tan\theta$

Now h_2 would be negative, so the surface meets the base at some distance $b < a$ from the lower wall, and the fluid's cross-section is a right-angled triangle with legs b along the base and h up the lower wall. The surface slope gives $h = b\tan\theta$, and the area gives $\frac{1}{2}bh = ka^2$. So $b^2\tan\theta = 2ka^2$, and

$$b = a\sqrt{\frac{2k}{\tan\theta}}, \qquad h = a\sqrt{2k\tan\theta}.$$

(Indeed $b < a$ precisely because $2k < \tan\theta$; the container is tall, so h causes no trouble.) The centroid of the triangle is a third of the way along each leg, so

$$\frac{x}{a} = \frac{1}{3}\sqrt{\frac{2k}{\tan\theta}} = \frac{1}{3}\sqrt{2k\cot\theta}, \qquad \frac{y}{a} = \frac{1}{3}\sqrt{2k\tan\theta}.$$

At $k = \frac{1}{2}\tan\theta$ both pairs of formulae give $x/a = \frac{1}{3}$ and $y/a = \frac{1}{3}\tan\theta$, so the two cases join up.

Toppling

Ask yourself What decides whether the container tips over the moment it is let go?

The container is light, so the only weight is that of the fluid, acting vertically through its centre of mass G. The plane can push only on the base, so the normal reaction acts somewhere on the base, between the lower edge O and the upper edge. If the vertical through G meets the plane downhill of O, the weight has a moment about O that no such reaction can balance, and the container rotates about O: it topples.

Take O as origin, with x up the slope and y perpendicular to it. The vertical direction downwards has components $(-\sin\theta, -\cos\theta)$ in these axes, so going down the vertical from $G = (x, y)$ through a distance $y\sec\theta$ reaches the plane at

$$x - y\tan\theta.$$

So the container topples if $x < y\tan\theta$.

Suppose $k < \frac{1}{2}$ and $\theta > 45°$. Then $\tan\theta > 1 > 2k$, so $k < \frac{1}{2}\tan\theta$ and we are in the triangular case. There

$$y\tan\theta - x = \frac{a}{3}\left(\sqrt{2k\tan\theta}\,\tan\theta - \sqrt{\frac{2k}{\tan\theta}}\right) = \frac{a}{3}\sqrt{\frac{2k}{\tan\theta}}\left(\tan^2\theta - 1\right) > 0,$$

because $\tan\theta > 1$. Hence $x < y\tan\theta$: the vertical through G passes below the lower edge, and the container topples, whatever the value of $k < \frac{1}{2}$. (In the triangular case the same calculation shows the container is stable if $\theta < 45^\circ$, so 45° is exactly the threshold there.) This is the result. The condition $k < \frac{1}{2}$ is what guarantees the triangular case when $\theta > 45^\circ$.

Checks

- With $\theta = 0$ the first case gives $x = \frac{1}{2}a$ and $y = \frac{1}{2}ka$, the centre of a block of fluid of depth ka.
- The two cases agree at $k = \frac{1}{2}\tan\theta$, where the trapezium degenerates into the triangle.
- A numerical check: integrating the fluid region directly for 300 random pairs (k, θ) matched both pairs of formulae to within 1.4×10^{-6} (with $a = 1$). For example, $\theta = 55^\circ$, $k = 0.3$ gives $x \approx 0.216a$, $y \approx 0.309a$, and $y\tan\theta \approx 0.441a > x$, so it topples.

Common slips

- Drawing the surface parallel to the base, or sloping the wrong way: the fluid piles up against the *lower* wall.
- Forgetting that the container is light, and adding a weight for the container: its centre of mass is that of the fluid.
- Testing the vertical through G against the centre of the base rather than against the lower edge O.

Beyond STEP

The test "does the vertical through the centre of mass meet the base?" is the simplest case of a stability analysis. Floating ships are treated the same way, but there the buoyancy acts through the centre of mass of the displaced water, which moves as the ship heels; this leads to the idea of the metacentre. Moving fluid (sloshing after release) is a much harder problem in fluid dynamics. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers rigid-body equilibrium and rotation about an axis carefully.

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